

D1 - PROJECT PROPOSAL

AgriUAV Platform: Precision Drone Dispatching Platform in the Mekong Delta

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1. Problem Statement

Agricultural cooperatives and drone service enterprises in the Mekong Delta currently face a critical lack of digital tools to schedule, dispatch, and monitor spraying drone fleets, resulting in the following measurable consequences:

High Operational Costs: Managing spraying schedules manually via Zalo, Excel, or paper notebooks causes schedule conflicts, overlapping shifts, and wasted deadhead flights, inflating operational costs by 15–20% (Nguyen et al., 2025; N=940 farms).

Low Field-Level Efficiency: Without accurate field mapping, drones spray uniformly across the entire field rather than concentrating variable dosages on specific pest-infected zones, resulting in excessive agrochemical wastage.

Safety Risks & Heavy Losses: The absence of automated safety geofencing has led to incidents where drones drifted off-course, causing neighbor crop poisoning, or crashed into 110kV high-voltage power lines (e.g., the Long An incident causing blackouts for 76,000 customers - Tuoi Tre, 2024).

Compliance Hardships: No automated flight logging tools exist, failing to satisfy mandatory reporting requirements regarding drone identification numbers and pilot certifications specified under Decree 288/2025/ND-CP (effective November 5, 2025).

1.1. Proposed Solution

The AgriUAV Platform is designed as a Web and Mobile application ecosystem to automate scheduling, optimize drone dispatching via digital field mapping, calculate chemical dosages automatically, and track spraying progress in real-time.

1.2. System Scope

Operation management of a drone fleet scaled up to a maximum of 50 drones per cooperative within the rice cultivation regions of the Mekong Delta. The system excludes direct hardware flight control interception and embedded computer vision AI for pest detection.

1.3. In-Scope - System Functions

- Drone fleet management: drone identification codes, model specifications, tank capacity, and operational status (Available / In-Flight / Under Maintenance)
- Pilot profile management: certifications, licenses, and flight history records compliant with Decree 288/2025/ND-CP
- Spray scheduling: farmers and HTX managers submit spray requests; system confirms and records assignments
- Automated drone dispatch: assignment logic based on battery level, current GPS position, and schedule availability
- Real-time spray progress tracking on digital field map: drone position overlay and percentage area completion
- Automated alerts: low pesticide level, low battery, adverse weather (wind speed ≥ 3 m/s), and geofencing boundary breach (including 50m buffer zone around registered high-voltage power lines)
- Pesticide inventory management and spray formula records per approved chemical registry
- Spray log export and seasonal compliance reporting for the Plant Protection Department
- Role-based access control: Admin / HTX Manager / Operator / Farmer
- System scale: maximum 50 drones per cooperative, within rice cultivation zones in the Mekong Delta

1.4. Out-of-Scope - Explicitly Excluded

- Direct drone hardware control from software (fly-by-wire or autopilot command interfaces)
- AI-based pest detection from drone imagery (computer vision or deep learning models)
- Integration with irrigation systems or other agricultural machinery
- Financial management, accounting, or VAT invoicing modules
- Online flight permit application with the Operations Department (UTM system integration)
- Physical cloud deployment or real-hardware drone integration

2. Business Case & ROI Analysis

2.1. Ecosystem Analysis - 3 Layer Model

To demonstrate the market gap that AgriUAV Platform will fill, the domain structure is categorized into three explicit layers:

Ecosystem Layer	Current Market State	AgriUAV Platform Intervention
Hardware Layer	Highly developed and saturated - DJI Agras (~80% Vietnam market share), XAG (~15%), YANMAR, and domestic brands (~5%), equipped with RTK positioning and multispectral cameras. Hardware availability is not a bottleneck.	Agnostic integration via mobile GPS telemetry at MVP stage. Planned native MAVLink telemetry protocol integration in future versions.
Service Layer	Local cooperatives (HTX) and private entities provide bulk drone spraying services (~150,000 VND/ha). Internal operations remain purely manual - managed via Zalo, Excel, and paper logbooks.	Provides the end-to-end digital coordination layer to automate scheduling, dispatch, and compliance for existing service providers.
Platform Layer	CURRENTLY VACANT - Total absence of a localized IS to optimize fleet dispatching, manage pilots, schedule trips, and maintain regulatory compliance logs. Confirmed by Thanh Nien (June 2025): 1,215 active drones across 3 provinces with no platform-layer management software reported.	AgriUAV Platform fills this strategic gap, serving as the enterprise-grade operational coordination system for drone service cooperatives.

2.2. Market Sizing (TAM / SAM / SOM)

Based on verified field statistics as of June 2025 and empirical research, market sizing is grounded in actual drone counts rather than projections. Note: the SOM figure is a projected estimate based on the Can Tho University - HTX extension partnership network; it is not a signed commitment and should be treated as a planning assumption.

Segment	Drone Count & Data Source	Target Customer	Strategic Rationale
TAM (Total Addressable Market)	~4,000 agricultural drones active across 1.5 million ha of rice cultivation in the Mekong Delta. <i>Source: Vietnam News (2026); Nguyen et al. (2025)</i>	Full Mekong Delta drone fleet	No digital platform in market = entire fleet is addressable
SAM (Serviceable Addressable Market)	1,215 active drones in 3 priority provinces: Kien Giang (690), An Giang (265), Dong Thap (260). <i>Source: Thanh Nien Newspaper, June 2025 (verified field statistics)</i>	HTX and commercial drone service operators in 3 core provinces	Operationally reachable in Year 1 rollout

SOM (Serviceable Obtainable Market)	200 drones managed by pilot-partner cooperatives linked to Can Tho University agricultural research network. <i>Note: this is a projected SOM based on the Can Tho University-HTX extension partnership network; not yet a signed contract. Conservative estimate subject to revision.</i>	Strategic pilot HTX partners	Direct partnership channel for initial deployment
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Revenue Model: B2B SaaS licensing at 500,000 VND / drone / month. Projected annual recurring revenue from SOM: $200 \times 500,000 \times 12 = 1,200,000,000$ VND/year.

2.3. Cost-Benefit Analysis (Reference: 1 HTX - 10 drones, 500 ha/season)

The following cost-benefit breakdown uses a representative mid-size cooperative as the reference unit. All AS-IS figures are team estimates derived from published operational benchmarks and the 15–20% cost inflation rate reported in Nguyen et al. (2025). Absolute VND values are conservative projections, not direct field measurements. Methodology notes are included in the rightmost column.

Cost Category	AS-IS Manual	TO-BE AgriUAV	Savings	Data Source / Methodology Note
Pesticide wastage (imprecise spraying)	~45M VND/season	~31.5M VND/season	–30% = 13.5M VND	Derived from DJI Agras T40 average application rate (11 L/ha) vs. precision dosing, applied to 500 ha $\times$ 150,000 VND/ha service cost benchmark. Conservative estimate; team derivation based on published operational benchmarks (Nguyen et al. 2025).
Drone fuel (duplicate flights, wrong zones)	~12M VND/season	~9.6M VND/season	–20% = 2.4M VND	Based on 15–20% deadhead flight waste rate from Nguyen et al. (2025); applied to estimated 10-drone fleet fuel cost at ~1.2M VND/drone/season.
Coordination & reporting labor	~8M VND/season	~2M VND/season	–75% = 6.0M VND	Estimated 4 admin-hours/week $\times$ 12 weeks $\times$ ~170,000 VND/hour = ~8.2M VND AS-IS. Platform automates >75% of

				<i>scheduling and report generation. Team estimate; no direct external source.</i>
Legal penalty risk (Decree 288 non-compliance)	Unquantified - ongoing exposure	Significantly reduced	Compliance built-in	<i>Decree 288/2025/ND-CP Article 23: administrative fines for non-compliant flight logging. Exact penalty amount not publicly specified; risk is qualitative.</i>
TOTAL ESTIMATED SAVINGS / SEASON	—	—	~21.9M VND	Sum of above quantified rows.

Platform operating cost estimate: ~1,200,000 VND/month (cloud infrastructure + API services). For a 3-month rice season, platform cost = 3,600,000 VND vs estimated savings of ~21,900,000 VND → **Positive ROI** from the first season of deployment.

Key performance improvements (summary):

- Operational cost reduction: 15–20% per season - validated by Nguyen et al. (2025)
- Compliance report turnaround: from 3–5 business days (manual) to under 1 hour (automated PDF/Excel export)
- Critical safety events: proactive geofencing blocks drone entry into registered high-voltage transmission line corridors
- Operator activity verification: GPS-based field check-in/out enforces verified physical presence

3. Stakeholder Analysis

The system serves stakeholders across three tiers. Two actors marked (*) were supplemented by the team after identifying gaps in the initial AI-generated stakeholder list, verified against Decree 288/2025/ND-CP and cooperative supply chain domain knowledge.

Classification	Stakeholder	Core Role & Expectations	Power	Interest	Strategic System Action
Primary	Cooperative Management (HTX)	Optimize fleet dispatching, lower operational costs by 15–20%, eliminate unverified pilot activity and pesticide leakage.	5	5	Intuitive Web Dashboard for scheduling, shift allocation, and compliance reporting.
Primary	Operators / Drone Pilots	Receive clear flight orders, view digital field boundaries, navigate flight paths safely away from power lines, and log verified flight hours.	2	5	Lightweight Mobile App with GPS navigation and automated field check-in/check-out logging.
Primary	Rice Farmers	Easily book a spray service, track real-time drone completion status, and verify actual pesticide usage. Avg. age ~51 (Nguyen et al. 2025), limited IT experience.	3	4	Automated SMS/Zalo notifications updating order status in real-time.
Primary	Agricultural Engineers / Agronomists	Define and validate spray formulas per crop type, manage approved pesticide registry, and confirm correct dosage application.	3	4	Formula management module with pesticide registry integration per Plant Protection Dept approved chemical list.
Secondary	Agricultural Exporters (Trung An, Minh Phu)	Trace pesticide application history data to meet strict clean grain export standards (EU, US) - VietGAP/GlobalGAP.	3	4	Exportable flight log modules matching VietGAP/GlobalGAP data standards.
Secondary	Drone Manufacturers (DJI, XAG)	Expand market reach through platform compatibility; benefit from telemetry integration via MAVLink SDK.	2	3	MAVLink telemetry API integration roadmap (planned post-MVP).
Secondary	Banks / Agricultural Credit Institutions	Use verified seasonal production data as a basis for agricultural lending risk assessment.	2	2	Seasonal operation summary export with verified flight logs usable as credit evidence.

Secondary (*)	Pesticide Retailers	Supply approved chemical inputs to HTX; inventory sync prevents unapproved pesticide use. [Supplemented by team - not in initial AI output]	2	2	Chemical inventory sync API connected to approved chemical registry.
Regulatory	Plant Protection Dept. (Cục Trồng trọt & BVTV)	Monitor compliance and check approved pesticide databases under Decree 288/2025.	5	3	Automated report generation (PDF/Excel) linking drone serial numbers and pilot licenses.
Regulatory (*)	Operations Dept. / General Staff (Cục Tác chiến)	Primary military authority for UAV flight permits and drone ID registration under Decree 288/2025/ND-CP. [Supplemented by team, confirmed via Vietnam Naval News 2024]	4	2	Store and display device tracking UUIDs and certified pilot license numbers in compliance reports.
Regulatory	Provincial Authorities (UBND / Sở NN&PTNT ĐBSCL)	Grant operational licenses for drone service providers at provincial level; receive periodic area-sprayed reports.	3	3	Periodic provincial spraying area reports exportable to Ministry of Agriculture.

4. AS-IS Process Analysis

The current manual operational workflow is broken down into 5 sequential phases. Field evidence and documented incidents expose critical pain points at each stage that require digital infrastructure solutions.

#	Phase	AS-Is Manual Workflow	Measurable Pain Points
1	Demand Capture	Farmers call or text the cooperative via Zalo to declare field areas needing spraying.	Ambiguous Boundaries: Lack of GPS coordinates leads to spraying wrong fields, causing accidental neighbor crop exposure and boundary disputes.
2	Scheduling & Assignment	Management writes details in a paper journal or logs them in a shared Excel file, then calls pilots individually.	Double-Booking & Deadhead Flights: Scheduling conflicts cause overlapping shifts and misallocated dispatch routes, inflating operational costs by 15–20% (Nguyen et al. 2025).
3	Field Execution	Pilots bring the drone to the field and spray based on intuition or standalone hardware controls with no centralized telemetry.	Unverified Operator Activity: Operators may under-dose chemical applications or fabricate flight records to claim fuel and chemical allocations without detection, no audit trail exists.
4	Monitoring & Control	No active tracking infrastructure exists. Management waits at the office until the pilot verbally confirms completion by phone.	CRITICAL SAFETY RISK: Drones collide with high-voltage power lines due to lack of geofencing (Long An 110kV incident, Tuoi Tre 2024 - 76,000 customers affected), or drift off-course, poisoning nearby plots.
5	Reconciliation & Reporting	At month-end, the cooperative tallies paper logs to invoice farmers and generates handwritten government reports.	Compliance Failure: Error-prone manual data cannot comply with Decree 288/2025/ND-CP, which mandates digital logs of drone IDs and pilot credentials. No verifiable audit trail exists.

5. Work Breakdown Structure (WBS)

WBS	Deliverable / Task	Member	Week	Specific Output
1.0	D1 - Project Proposal	All	W1-W3	Complete proposal file
1.1	Choose topic, analyze the UAV domain	BA + PM	W1	Preliminary list of groups and topics
1.2	Stakeholder mapping + AS-IS analysis	BA	W2	Stakeholder map + Business case draft
1.3	Business case, cost-benefit, ROI	BA + PM	W3	Business case with specific data
1.4	WBS + Timeline + Risk Register + Gantt Chart	PM	W3	WBS table, Gantt chart (Section 6.1), risk log
1.5	Internal peer review + completion of D1	All	W3	Submit D1
2.0	D2 - SRS Document	BA + SA	W4	SRS following IEEE 830
2.1	Role-play interview stakeholders	BA	W4	Interview notes
2.2	Elicit & classify ≥ 16 FR + ≥ 7 NFR	BA + SA	W4	FR/NFR table with priority ratings
2.3	Write ≥ 10 Use Cases + User Stories	SA	W4	UC description + acceptance criteria
2.4	Complete SRS according to IEEE 830	BA + SA	W4	Submit D2
3.0	D3 - System Modeling Package	SA	W5-W6	DFD + UML + ERD
3.1	DFD Level 0 + Level 1	SA	W5	Balanced DFD diagrams
3.2	Class Diagram + ≥ 3 Sequence Diagrams	SA	W5-W6	UML diagrams in StarUML
3.3	ERD 3NF + Data Dictionary	SA + SD	W6	Logical + Physical ERD
3.4	Traceability matrix SRS $\leftrightarrow$ Models	SA	W6	Submit D3
4.0	D4 - System Design Document	SD	W7-W9	Complete SDD
4.1	UI Wireframes ≥ 5 screens (Figma)	SD + QA	W7	Mobile app + web dashboard
4.2	Architecture + 2 alternatives + trade-off	SD	W8	Layered vs Microservices analysis
4.3	Physical data schema + security model	SD	W9	PostGIS schema + threat model
4.4	QA plan + test cases + ops plan	QA	W9	Submit D4
5.0	D5 - Final Presentation & Defense	QA + All	W10	Slide + Demo + Q&A

6. Project Timeline

Week	Phase	Main Activities	Deliverable	Milestone
W1	P1 - Business	Kickoff, domain selection, team formation, stakeholder screening.	Topic proposal	Groups + domains confirmed
W2	P1 - Business	Stakeholder map, AS-IS analysis, pain points.	Business case draft	Complete Stakeholder Map
W3	P1 - Business	Business case, WBS, Risk Register, Gantt chart, peer review.	D1 Proposal	SUBMIT D1 - end of W3
W4	P2 - Requirements	Interview, elicit FR/NFR, write UC + SRS.	D2 SRS	SUBMIT D2 - end of W4
W5	P3 - Modeling	PT1, DFD L0+L1, Class Diagram draft.	PT1 + DFD draft	Progress Test 1
W6	P3 - Modeling	Complete UML, ERD, and traceability matrix.	D3 Models	SUBMIT D3 - end of W6
W7	P4 - Design	UI wireframes Figma, persona, user journey.	UI wireframes	≥5 complete screens
W8	P4 - Design	PT2, Architecture pattern, tech stack.	PT2 + Architecture	Progress Test 2
W9	P4 - Design	Trade-off analysis, security model, QA plan, ops.	D4 SDD	SUBMIT D4 - end of W9
W10	P5 - Defense	PT3, mock defense, final presentation.	D5 Defense	FINAL PRESENTATION - W10

6.1. Gantt Chart – 10 Week Project Schedule

The chart below maps all deliverables and progress tests to their target weeks. Each color represents a distinct deliverable. Milestone weeks (D-submit weeks) are shaded green in the timeline table above.

Activity / Deliverable	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
D1 - Project Proposal										
D2 - SRS (IEEE 830)										
PT1 - Progress Test 1										
D3 - System Modeling										
D4 - System Design										
PT2 - Progress Test 2										

PT3 - Progress Test 3										
D5 - Final Defense										

Legend: Blue = D1 Proposal | Green = D2 SRS | Orange = Progress Tests (PT1/PT2/PT3) | Yellow = D3 System Modeling | Purple = D4 System Design | Red = D5 Final Defense

7. Risk Register

#	Risk	Probability	Impact	Level	Mitigation
R1	Scope creep extending too many modules	High	High	Critical	Keep the out-of-scope frozen; each new feature requires PM approval before inclusion.
R2	Lack of factual data from cooperatives/farmers	Medium	High	High	Use data from Cục Trồng trọt reports, Nguyen et al. (2025), and verified press reporting as substitutes.
R3	Member takes unexpected leave mid-term	Low	High	High	Fully documented tasks on Notion; each deliverable has a designated backup member.
R4	Inconsistency between SRS, UML, SDD	High	High	Critical	Traceability matrix updated and reviewed after each deliverable submission.
R5	Lack of time to complete D4 (Week 9)	Medium	Medium	Medium	Begin architecture design from W7; do not allow work to accumulate into W9.
R6	Legal basis of Decree 288/2025 is unclear, making compliance module design difficult	Medium	Low	Medium	Scope compliance module at PDF report output level - not directly integrated with Operations Dept.
R7	Regulation change risk - ND 288/2025 may be supplemented by additional decrees	Low	Low	Low	Record assumption: 'Design is based on ND 288/2025 as of analysis date - requires review if new decrees are issued.'
R8	Domain knowledge gap team lacks direct field experience in ĐBSCL	Medium	Medium	Medium	Compensate via peer-reviewed literature (Nguyen et al. 2025), field reporting, and role-play stakeholder interviews in W4.

8. Team Roles & Responsibilities

Member	Role	Main Deliverable	Notes
Dang Thanh Cong	Project Manager - planning, progress tracking, team coordination	Summary of all deliverables	Contact the instructor; maintain traceability matrix
Nguyen Thi Thanh Thao	Business Analyst - business analysis, stakeholder mapping, requirements elicitation	D1 Proposal, D2 SRS	Lead stakeholder interviews and FR/NFR elicitation
Le Van Minh	System Analyst - DFD, UML, ERD, and traceability models	D3 System Models	Use PlantUML for all diagrams
Tran Anh Khoa	System Designer - architecture, UI/UX Figma, data design, security	D4 SDD	Use Figma for wireframes; PostGIS for data schema
Dang Vo Thanh Hieu	Presenter / QA - slides, storytelling, quality control, Q&A preparation	D5 Presentation	Cross-check consistency across all deliverables

9. D1 Quality Checklist

Criteria	Completion	Reference
Stakeholders mapped (Primary / Secondary / Regulatory) with Power & Interest ratings	Complete	Section 3
Measurable problem statement with quantified impact (15–20% cost, 76,000 affected customers)	Complete	Section 1
Clear scope - In-scope (10 functions) and Out-of-scope (6 exclusions) explicitly listed	Complete	Section 1.3–1.4
Business case with specific data - 3-layer ecosystem, TAM/SAM/SOM, cost-benefit table, ROI	Complete	Section 2
WBS + Gantt chart + Timeline + Risk Register (≥5 risks with likelihood, impact, mitigation)	Complete	Sections 5, 6, 6.1, 7
AI Audit Log entries #01-08 included (W1-W3 phase coverage)	Complete	Section 10
Cost-benefit data source notes added for all quantitative figures	Complete	Section 2.3
SOM methodology disclaimer added	Complete	Section 2.2
R8 probability corrected (Low → Medium) with rationale note	Complete	Section 7

10. AI Artifact Audit Log - Entries #01-08 (W1-W3)

This consolidated audit log covers all AI interactions for Phase 1 (Problem & Business). Entries #01–#03 are from W1, #04–#06 from W2, and #07-08 from W3. The cumulative log continues in D2 (entries #09–#12) and will reach ≥25 entries by end of W10.

No.	Week	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta / Verification
01	W1 / P1	Problem & Business Decomposition	Claude	"We are developing an IS project for ITA301 on Agricultural UAVs in the Mekong Delta. Suggest 3 distinct sub-domain directions, stating the primary business problem and implementation complexity for each."	3 directions: (1) Fleet Management generic tracking; (2) Precision Spraying Platform localized scheduling; (3) Marketplace hourly drone rentals. All rated 3 star complexity.	SELECTED (2) Precision Spraying Platform: Empirical data: Nguyen et al. 2025, N=940. Pain points validated: HTX uses Zalo/Excel. Platform layer is vacant in ecosystem. REJECTED (3): Scope too broad, no IS depth. REJECTED (1): Overlaps with another team's confirmed topic. Verification: Ecosystem gap confirmed - Lao Dong (2025) confirms digital scheduling problem remains unresolved in DBSCL cooperatives.
02	W1 / P1	Pattern Recognition & Comparison	ChatGPT	"Compare Agricultural UAV, Delivery UAV, and Surveillance UAV across: (a) stakeholder complexity, (b) regulatory burden in Vietnam, and (c) data availability for an IS academic project."	AI concluded Delivery UAV is 'easiest' due to lower perceived regulation. Agricultural UAV ranked medium complexity.	AI RANKING REJECTED incorrect evaluation: Agricultural UAV has a more structured legal framework (Decree 288/2025/ND-CP + Plant Protection Dept) than Delivery UAV. Maintained Agricultural UAV: (a) peer-reviewed research available (Nguyen 2025), (b) verifiable field incidents, (c) richer stakeholder map for IS analysis. Verification: Decree 288/2025/ND-CP cross-checked via thuvienphapluat.vn confirms specific drone ID and pilot certification logging requirements.
03	W1 / P1	Decomposition - Ecosystem Mapping	Claude	"For the AgriUAV Platform (ITA301, Mekong Delta): decompose the current	AI proposed a 2-layer split: Hardware/Device	REJECTED 2-layer model oversimplified and inaccurate.

				Agricultural UAV ecosystem into structural layers. Identify which layers are occupied and which represent market gaps for an IS solution."	layer and Application/User layer. Suggested the gap lies in 'application integration.' No distinct Service Layer or Platform Layer identified.	Team defined 3-Layer Model: Hardware Layer (DJI/XAG occupied), Service Layer (HTX fleets occupied, manual), Platform Layer (digital IS coordination VACANT). Verification: 3-layer model validated by Thanh Nien (June 2025): 1,215 active drones across 3 provinces with no platform-layer management software reported.
04	W2 / P1	Decomposition - Stakeholder Mapping	Claude	"Project: AgriUAV Platform for drone spraying in Mekong Delta. List all stakeholders, classify as Primary/Secondary/Regulatory, and outline core roles and expectations for each."	Suggested 6 general groups: Farmers, HTX, Pilots, Drone Manufacturers (DJI), Software Developers, and Local Government totaling ~12 stakeholders.	5 core stakeholders accepted. Removed 'Software Developer Team' this is the project team, not an external stakeholder. SUPPLEMENTED 2 critical actors AI omitted: Operations Dept. / General Staff (sole authority for UAV flight permits under Decree 288/2025); Pesticide Retailers (essential for inventory/compliance integration). Split generic 'Government' into 2 specific regulatory bodies: Plant Protection Dept. + Operations Dept. Verification: Operations Dept. authority confirmed via Vietnam Naval News (2024).
05	W2 / P1	Decomposition - AS-IS Process Analysis	Claude	"Decompose the AS-IS business process of an agricultural cooperative in Vietnam executing drone spraying services. List each step, the actor responsible, and pain points at each stage. Context: currently managed via Zalo/Excel."	Proposed a basic 4-step generic workflow: Intake → Dispatch → Execution → Reporting. High-level pain points only: lack of tracking, slow report compilation.	Accepted 4-step structure as baseline. Expanded to 5 steps added 'Monitoring & Control' as a distinct phase (AI merged this into Execution). Localized pain points: Phase 3 operators may under-dose chemicals or inflate flight hours (AI missed this risk entirely); Phase 4 Long An 110kV power line collision

						(Dan Viet 2026) as concrete geofencing evidence; Phase 5 linked compliance failure directly to Decree 288/2025/ND-CP. Verification: 110kV incident: Tuoi Tre (2024) drone collision in Long An Province, 76,000 customer blackout.
06	W2 / P1	Process / Algorithm - Business Case	Claude	"Provide an industry-standard business case structure for an IS research project, including a preliminary ROI template and a formal problem statement formulation suitable for a university-level IS analysis project."	Standard 5-part structure (Context → Problem → Impact → Solution → Scope) and a generic theoretical ROI formula: (cost saved / investment) × 100%.	Adopted structural blueprint as foundation. Applied Mekong Delta data localization: 15–20% operational cost figure from Nguyen et al. (2025) N=940; 1,215 drones across 3 provinces from Thanh Nien June 2025. Added HTX Cost-Benefit table (not in AI output): pesticide –30%, fuel –20%, labor –75% = ~21.9M VND savings vs. 3.6M VND platform cost → positive ROI from Season 1. Note: absolute VND figures in cost-benefit table are team derivations from published operational benchmarks see methodology notes in Section 2.3. Verification: 15–20% cost: Nguyen et al. (2025). Province drone counts: Thanh Nien (June 2025).
07	W3 / P1	Pattern Recognition - Business Case Comparison	Claude	"Compare the business case of an Agricultural UAV platform vs. a Delivery UAV platform in Vietnam: market size, competition level, scalability, and suitability for IS research."	AI concluded Delivery UAV has 'a larger market' because logistics is booming. Agricultural UAV was rated lower on 'tech sophistication.'	COUNTER-ARGUMENT AGAINST AI BIAS: AI conflated 'market size = revenue potential' with 'research quality.' ITA301 evaluates domain richness, not revenue. Agricultural UAV is more suitable: (1) peer-reviewed paper exists (Nguyen et al. 2025, RWAE Journal) no equivalent for Delivery UAV in

						Vietnam; (2) real field data available by province from Vietnamese press; (3) richer stakeholder map; (4) 'Platform layer vacant' claim provable via 3-layer ecosystem decomposition. Team maintained Agricultural UAV selection. AI bias pattern recognized and documented. Verification: Pattern recognition team identified AI bias toward 'market = revenue' framing vs. ITA301 standard of 'domain richness = research quality.'
08	W3 / P1	Decomposition - WBS & Risk Register	Claude	"Propose a Risk Register for a 10-week IT project analyzing agricultural drone spraying in ĐBSCL. List 8–10 risks with likelihood (H/M/L), impact, and mitigation strategy."	AI listed 10 risks: scope creep, timeline, team conflict, data quality, tool learning curve, stakeholder availability, budget, integration complexity, security breach, deployment failure.	Accepted 5 generic risks (scope creep, timeline, team conflict, stakeholder availability, data quality). REMOVED 'deployment failure' ITA301 does not require actual deployment. REMOVED 'budget' not applicable to a university project. SUPPLEMENTED 4 UAV-domain-specific risks AI missed: Regulation change risk (NĐ 288/2025 newly effective, supplementary decrees possible); Domain knowledge gap (team lacks direct field experience probability set to Medium, corrected from Low); Literature availability; Stakeholder interview difficulty. Verification: NĐ 288/2025 confirmed effective via VnBusiness and Government E-Portal.

11. Verified Academic & Regulatory References

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